



Tema 3: Librerías/Componentes de Spark

Rafael Pastor Vargas
Andrés Duque Fernández
SCC-ETSI Informática-UNED

LIBRERÍAS/COMPONENTES DE SPARK

- Librerías disponibles en Spark.
- Gestión de datos al modo SQL: Spark SQL.
- Machine Learning en Spark: Spark MLlib.
- Procesamiento de gráfos: Spark GraphX.
- **Introducción a Spark Streaming.**

SPARK STREAMING

- Spark Streaming es la librería de procesamiento de streams de datos
- Escalable, alto rendimiento, tolerante a fallos, batch-oriented
- Streams de datos continuos discretizados (DStreams) of continuous data
- Integración con otros módulos/librerías
  - Spark SQL / Mllib/ML Pipelines



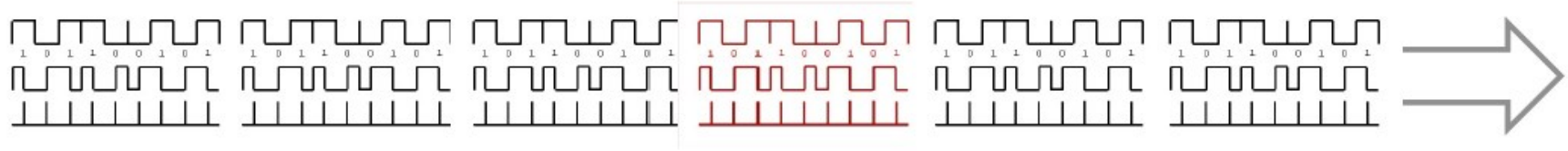


CASOS DE APLICACIÓN

Fraud detection in bank transactions



Anomalies in sensor data



Cat videos in tweets



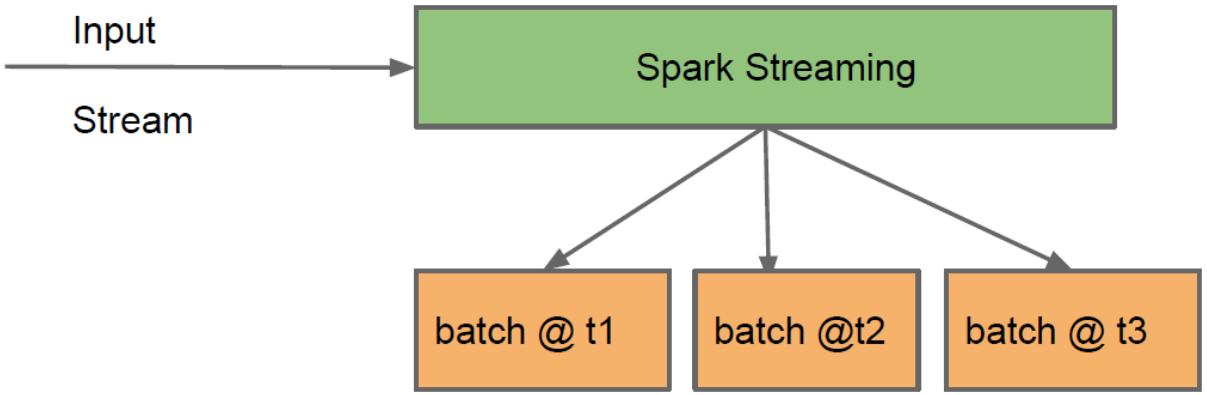
STREAMINGCONTEXT

- StreamingContext es el punto de entrada para la creación de aplicaciones basadas en Streams de datos

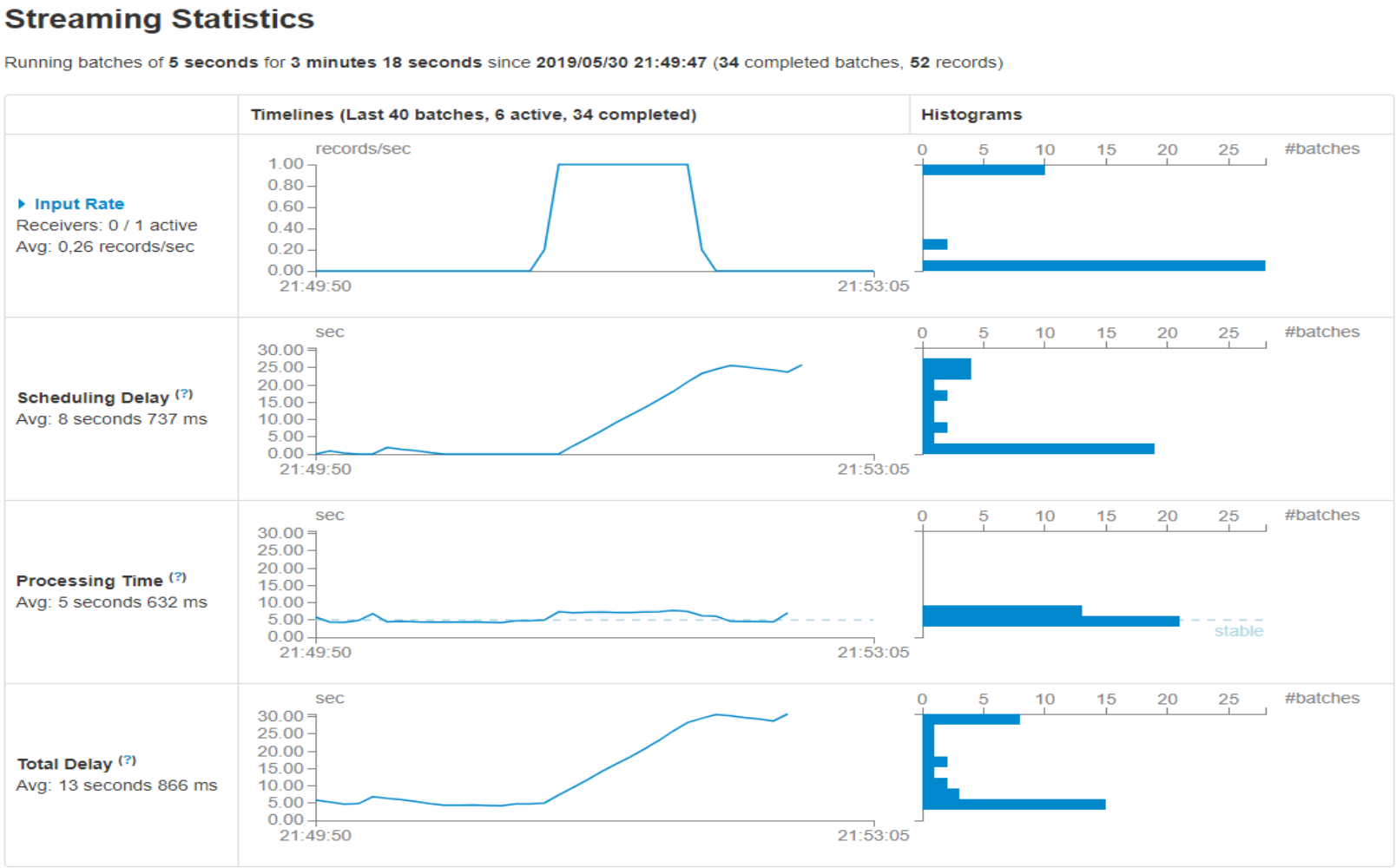
  val ssc = new StreamingContext(sc, BATCH_INTERVAL)

  scc = StreamingContext(sc, BATCH_INTERVAL)

- Se usa para crear los Dstreams (abstracción que representa el flujo de datos entrante)



STREAMING STATISTICS PAGE (1/2)



STREAMING STATISTICS PAGE (2/2)

▾ Active Batches (6)

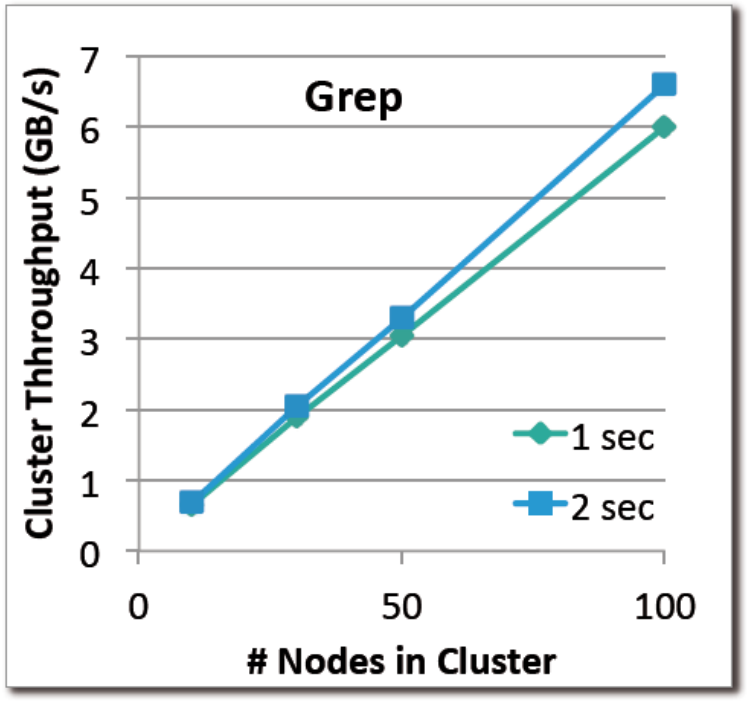
Batch Time	Records	Scheduling Delay (?)	Processing Time (?)	Output Ops: Succeeded/Total	Status
2019/05/30 21:53:05	0 records	-	-	0/1 (1 running)	queued
2019/05/30 21:53:00	0 records	-	-	0/1 (1 running)	queued
2019/05/30 21:52:55	0 records	-	-	0/1 (1 running)	queued
2019/05/30 21:52:50	0 records	-	-	0/1 (1 running)	queued
2019/05/30 21:52:45	0 records	-	-	0/1 (1 running)	queued
2019/05/30 21:52:40	0 records	26 s	-	0/1 (1 running)	processing

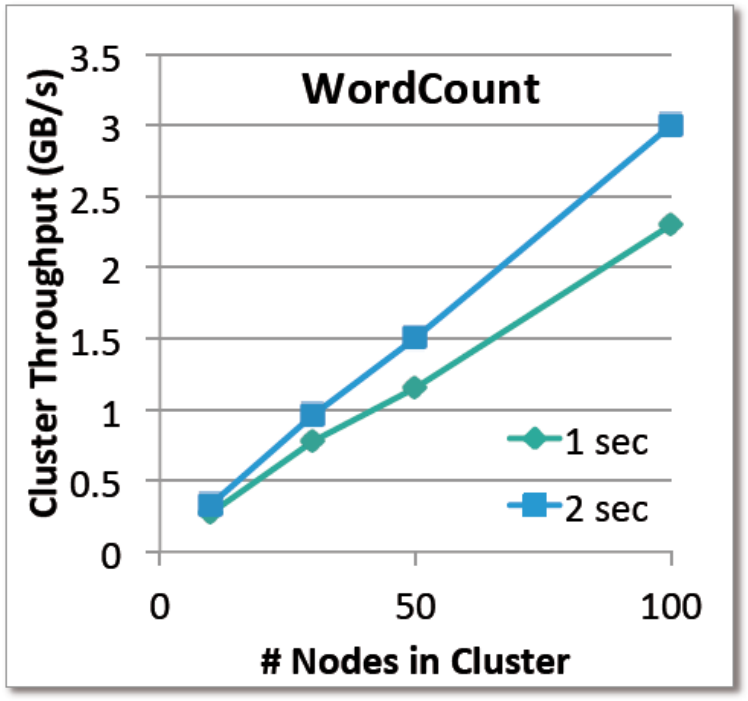
▾ Completed Batches (last 34 out of 34)

Batch Time	Records	Scheduling Delay (?)	Processing Time (?)	Total Delay (?)	Output Ops: Succeeded/Total
2019/05/30 21:52:35	0 records	24 s	7 s	31 s	1/1
2019/05/30 21:52:30	0 records	24 s	4 s	29 s	1/1
2019/05/30 21:52:25	0 records	25 s	5 s	29 s	1/1
2019/05/30 21:52:20	0 records	25 s	5 s	30 s	1/1
2019/05/30 21:52:15	0 records	26 s	5 s	30 s	1/1
2019/05/30 21:52:10	0 records	25 s	6 s	31 s	1/1
2019/05/30 21:52:05	1 records	23 s	6 s	30 s	1/1
2019/05/30 21:52:00	5 records	21 s	7 s	28 s	1/1
2019/05/30 21:51:55	5 records	18 s	8 s	26 s	1/1
2019/05/30 21:51:50	5 records	16 s	7 s	23 s	1/1
2019/05/30 21:51:45	5 records	13 s	7 s	21 s	1/1
2019/05/30 21:51:40	5 records	11 s	7 s	18 s	1/1
2019/05/30 21:51:35	5 records	9 s	7 s	16 s	1/1

RENDIMIENTO

- Puede procesar 6 GB/s en un clúster con 100 nodos (latencia por debajo del segundo)





DOMINIOS DE APLICACIÓN

Spark Streaming Use Cases

Data Integration

- **Streaming ETL:** Continually clean and aggregate data before pushing it into data stores.
- **Data enrichment:** Enrich live data by combining it with static data.

Business Intelligence & Analytics

- **Complex Event Processing:** Combine data from multiple sources to infer events or patterns.
- **Real-time Alerts:** detect and respond to unusual behaviors ('trigger events') quickly in real-time.

Predictive Analytics & Machine Learning

- **Fraud Detection:** Analyze transactions request to identify that the request is legitimate or not.
- **Recommendation Systems:** Produce a list of recommendations based on user's past behavior.

- https://www.dezyre.com/article/top-5-apache-spark-use-cases/271